

DE&VC TUTORIAL SHEET-2 (ECE, CSE) A. YEAR:2025-2026 Sem-II

Long Answer Questions:

1.Solve $x(y - z)p + y(z - x)q = z(x - y)$.
2.Solve $(mz - ny)p + (nx - lz)q = ly - mx$
3.Solve $\frac{\partial^3 z}{\partial x^3} - 7 \frac{\partial^3 z}{\partial x \partial y^2} - 6 \frac{\partial^3 z}{\partial y^3} = e^{2x+y}$
4.Solve $(D^3 - 4D^2D' + 4DD'^2)z = 2 \sin(3x + 2y)$
5. Find the angle between the normals to the surface $x^2 = yz$ at the points (1, 1, 1) & (2, 4, 1).
6. Find the directional derivative of $\phi = xyz$ along the direction of the normal vector to the surface $f = x^2z + y^2x + yz^2 = 3$ the point (1, 1, 1).
7. S.T the vector $\vec{F} = (x^2 - yz)\vec{i} + (y^2 - zx)\vec{j} + (z^2 - xy)\vec{k}$ is irrotational and find its scalar potential.
8.S.T $\text{grad} \left[\text{div} \left(\frac{\vec{r}}{r} \right) \right] = \frac{-2\vec{r}}{r^3}$ where $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$ and $r = \vec{r} $
9. If $\vec{F} = x^2\vec{i} + y^2\vec{j}$ then evaluate $\int_C \vec{F} \cdot d\vec{r}$ where C is the arc of the parabola $y=x^2$ From (0, 0) to (1, 1).
10. Find the work done in moving particle in the force field $\vec{F} = 3x^2\vec{i} + (2xz - y)\vec{j} + z\vec{k}$ along the straight line from (0, 0, 0) to (1, 2, 3)
11. Verify Green's theorem for $\int [(3x^2 - 8y^2)dx + (4y - 6xy)dy]$ where c is the region bounded by $x=0, y=0, x+y=1$
12.State Greens theorem and Verify Green's theorem for $\int [(xy + y^2)dx + x^2dy]$, where c is bounded by $y = x$ and $y = x^2$

Short Answer Questions:

1.Solve $P\sqrt{x}+q\sqrt{y} = \sqrt{z}$
2.Solve $\frac{y^2z}{x}p + xzq = y^2$
3.Solve $\frac{\partial^3 u}{\partial x^3} + 2\frac{\partial^3 u}{\partial x^2 \partial y} - \frac{\partial^3 u}{\partial x \partial y^2} - 2\frac{\partial^3 u}{\partial y^3} = 0$
4. Define Gradient of a Scalar point function and find Grad ϕ where $\phi = \log(x^2 + y^2 + z^2)$ at (1,1,1)
5.Find the Unit normal vector to the surface $xy^3z^2 = 4$ at the point (-1, -1, 2).
6. If $A = 3xyz^2\bar{i} + 2xy^3\bar{j} - x^2yz\bar{k}$ find $(A \cdot \nabla)\phi$ at (1, -1,1) and $\phi = 3x^2 - yz$
7.Is the vector $f = (x^2 + xy^2)\bar{i} + (y^2 + x^2y)\bar{j}$ irrotational?
8. Define Divergence and curl of a vector and find $\text{div } \bar{F}$ and $\text{curl } \bar{F}$ where $\bar{F} = \text{grad}(x^3 + y^3 + z^3 - 3xyz)$
9.If $\bar{F} = (x + 3y)\bar{i} + (y - 2z)\bar{j} - (x + pz)\bar{k}$ is solenoidal Find p
10.If $\bar{a}$ is a constant vector P.T $\text{Grad}(\bar{a} \cdot \bar{r}) = \bar{a}$ where $\bar{r} = xi+yj+zk$
11.In what direction from the point (1, -1,3) the directional derivative of $\phi=2xy+z^2$ is maximum and what is the magnitude of this maximum?
12. Evaluate $\int y^2dx - 2x^2dy$ along the parabola $y = x^2$ from (0,0) to (2,4).
13.State Green's theorem
14.State Stoke's theorem
15.State Gauss-Divergence theorem



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